FlexiPath – predictive, traffic-aware connectivity for moving users

Keep the session alive across Wi-Fi, 5G, LEO satellite and wired links by predicting the handover instead of reacting to the failure.



TEAM TAROT CLUB

Al Ain University

EDGE GROUP CHALLENGE

Advanced Technology Pioneers 2026

LIVE PROTOTYPE

flexipath.vercel.app

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▶ TRY THE LIVE PROTOTYPE
flexipath.vercel.app
guided demo, simulator and the 2-minute video

Executive summary

FlexiPath keeps critical sessions alive without the power penalty

PROBLEM

Today's stacks are break-before-make. A moving user loses the session, every TCP connection resets, and a drone whose command link goes silent for more than **300 ms** must stop and hover (3GPP TS 22.125 piloted-flight rule).

MECHANISM

FlexiPath predicts a fading link, wakes and validates the next modem just in time, duplicates only critical traffic for a **400 ms** burst, switches with a **0 ms** gap, and keeps every session alive through a fixed anchor relay.
Modems sleep in DRX the rest of the time.

EVIDENCE

160 paired simulated runs (scenarios A–H, 5 seeds, 4 speeds) against a legacy stack and two existing mechanisms (MPTCP/ATSSS reactive multipath and MPTCP-redundant/PDCP duplication).
FlexiPath kept the command link on **100 %** of flights in every scenario; the legacy stack on **0 %;** reactive multipath failed in three scenarios; permanent duplication draws **2.1–2.6×** the power.

DELIVERED

4–7 percentage points of critical delivery per watt for FlexiPath versus **0.3–0.9** for permanent duplication.

A working prototype (interactive 3D digital twin with a guided demo), a deterministic simulation with a headless benchmark, and a documented model with calibrated parameters.
All results are simulated; no live-network measurements are claimed.



The problem and why it matters



Networks hand over. Sessions don't. Legacy break-before-make costs **3.8 s** per wireless break.

Dropped calls. Frozen streams.
300 ms of silence, and a drone hovers.

Emergency response, industrial automation, connected transport and defence can't wait.

Two drones. One corridor. Dock → Wi-Fi 6 → 5G SA → LEO satellite. **C2, voice, video.**

Why a drone C2 mission: in EDGE's domains a silent command link is a safety event, and 3GPP TS 22.125 gives a published pass/fail rule for it (hover after 300 ms of silence).

1. DOCK (WIRED)
Stable baseline mission prep

2. WI-FI 6
Local coverage on the move

3. 5G SA
Wide-area coverage at speed

4. LEO SATELLITE
Global reach beyond line of sight

C2 LINK (CRITICAL)
Command and control

PUSH-TO-TALK VOICE (REAL-TIME)
Clear, low-latency communications

CAMERA UPLOADS (BULK)
High-throughput data transfer

Existing mechanisms we analysed (criterion 01)

REACTIVE MULTIPATH

MPTCP backup subflow (RFC 8684) and 3GPP ATSSS active-standby.

Anchored session with a standby path; moves only after loss is confirmed and the standby modem must wake.

0.35 s
gap that breaches the 300 ms C2 budget

1.20 W
low power consumption

ALWAYS DUPLICATE

MPTCP redundant scheduler and PDCP duplication.

Every protected packet is sent on two paths all the time.

99–100 %
critical delivery (most reliable)

2.65–3.39 W
high power consumption and 49–100 % duplicate protected bytes

Still fails when both paths die together (a shared radio shadow).

COMMON SHORTCOMINGS

Neither looks ahead.

No use of the signal trend.

No use of what is known about the route.

No distinction between a command packet and a video frame.



WHAT WE ADAPT

Keep the **anchored session** (from multipath).
Keep the **duplication** (from PDCP) but confine it to a **400 ms** burst around a predicted failure.
Add **prediction**, **route knowledge** and **traffic classes**.

References: RFC 8684, Multipath TCP v1 (IETF, 2020) · 3GPP TS 23.501 §5.32, ATSSS · 3GPP TS 38.323, PDCP duplication · 3GPP TS 22.125 §7.2, UAV C2 KPIs · full analysis: docs/existing-mechanisms.md

The FlexiPath mechanism

(criterion 02)



DETECT

Measure the primary link ten times a second.
Other modems sleep in DRX at 1.2 W and report signal once per cycle.

PREDICT

Extrapolate the signal trend to where packet loss reaches 15 %.
Arm the loop under 8 s; trigger under 3 s.
Coverage map warns 8 s ahead of a known shadow.

PREPARE

Wake the best alternate modem to a warm probe (1.8 W) and validate it with 64-byte heartbeats.
Measure RTT and loss before it carries anything.

REPLICATE

For a 400 ms burst (3.4 W) send critical traffic, and real-time traffic if its loss leaves budget, on both paths.
The receiver keeps the first copy of each sequence number and discards the late duplicate, so head-of-line delay is 0 ms.
Bulk traffic is never copied, only paced.

HONEST BEHAVIOUR

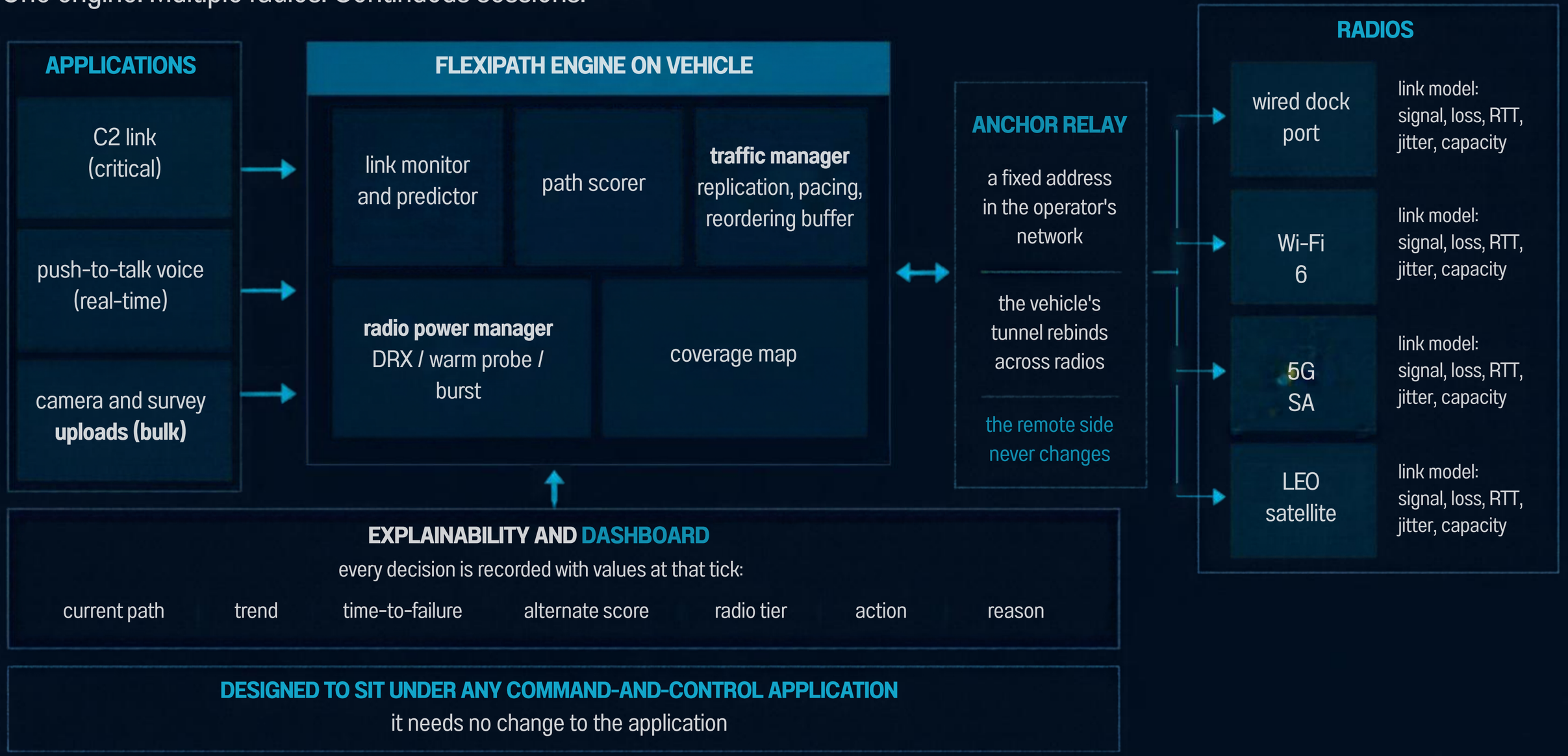
A trend that reverses releases the prediction at no cost.
An unpredictable outage falls back to a fast failover (160 ms in our model).

SWITCH and CONTINUE

Make the prepared path primary with a 0 ms gap and put the old radio back to sleep.
Sessions stay intact through a tunnel to a fixed anchor relay.

System design

One engine. Multiple radios. Continuous sessions.



Prototype, simulation and model (criterion 03)

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interactive 3D digital twin with mission and advanced views

guided demo

pauses at each real decision, explains what happened and which panel shows it

deterministic 10 Hz engine

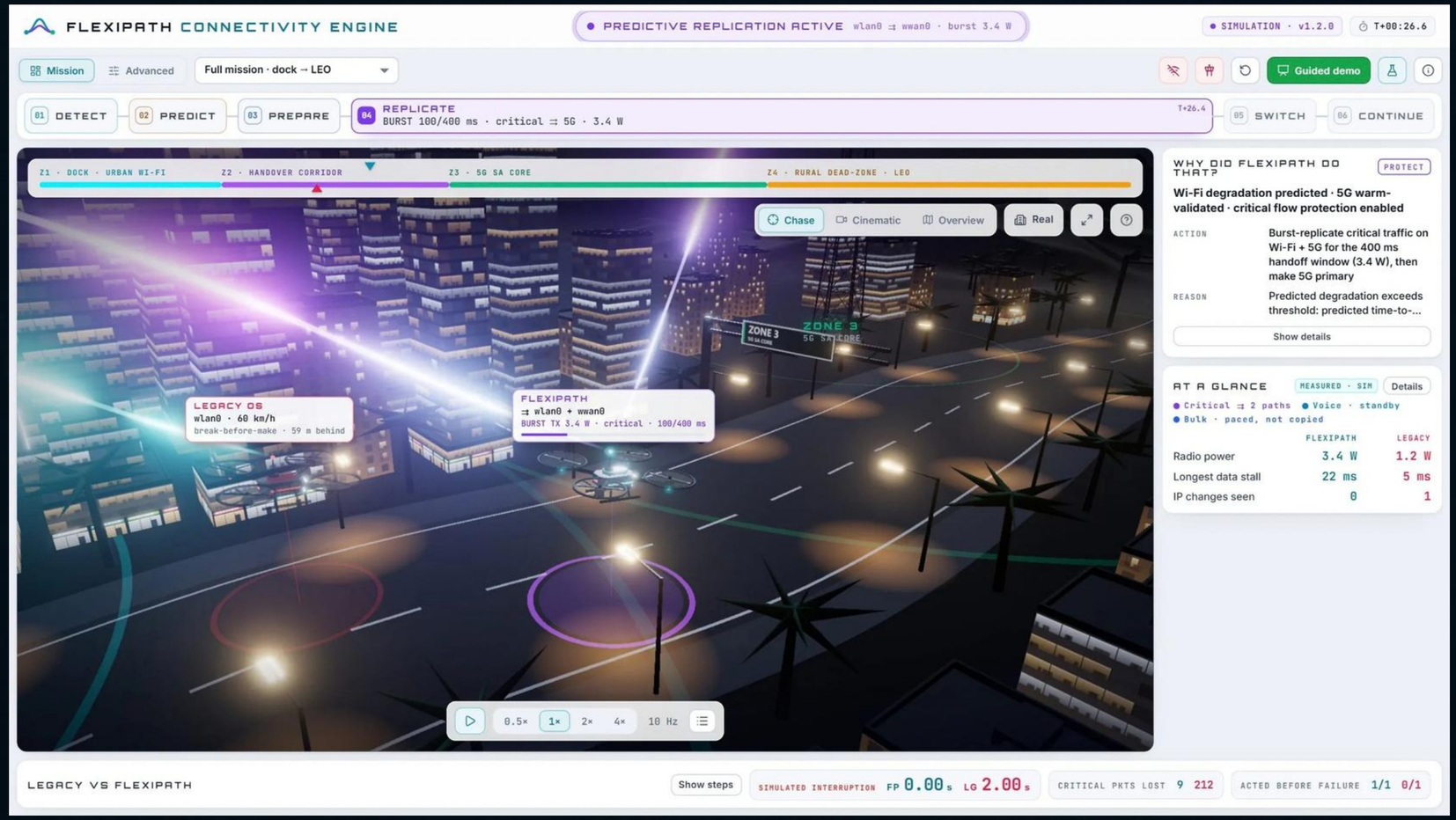
seeded random streams with fading, capacity, loss, RTT, jitter, TCP and radio models

headless benchmark

scenarios A–H × 5 seeds × 4 speeds = 160 paired runs per stack, plus baselines and ablations; exported as CSV with 95 % CIs

published model

parameters checked against published measurements; the 100 ms modem wake tested at 260 ms as a sensitivity run



Mission view of the live prototype: the FlexiPath drone (cyan) and the legacy drone (crimson) fly the same seed and radio conditions, with the decision loop, the decision card and the scoreboard around the twin.

▶ Try it yourself: **flexipath.vercel.app** · the guided demo, the simulator and the 2-minute video

Results against the existing mechanisms (criterion 04)

Paired means over 20 runs per scenario, same seeds for every stack.

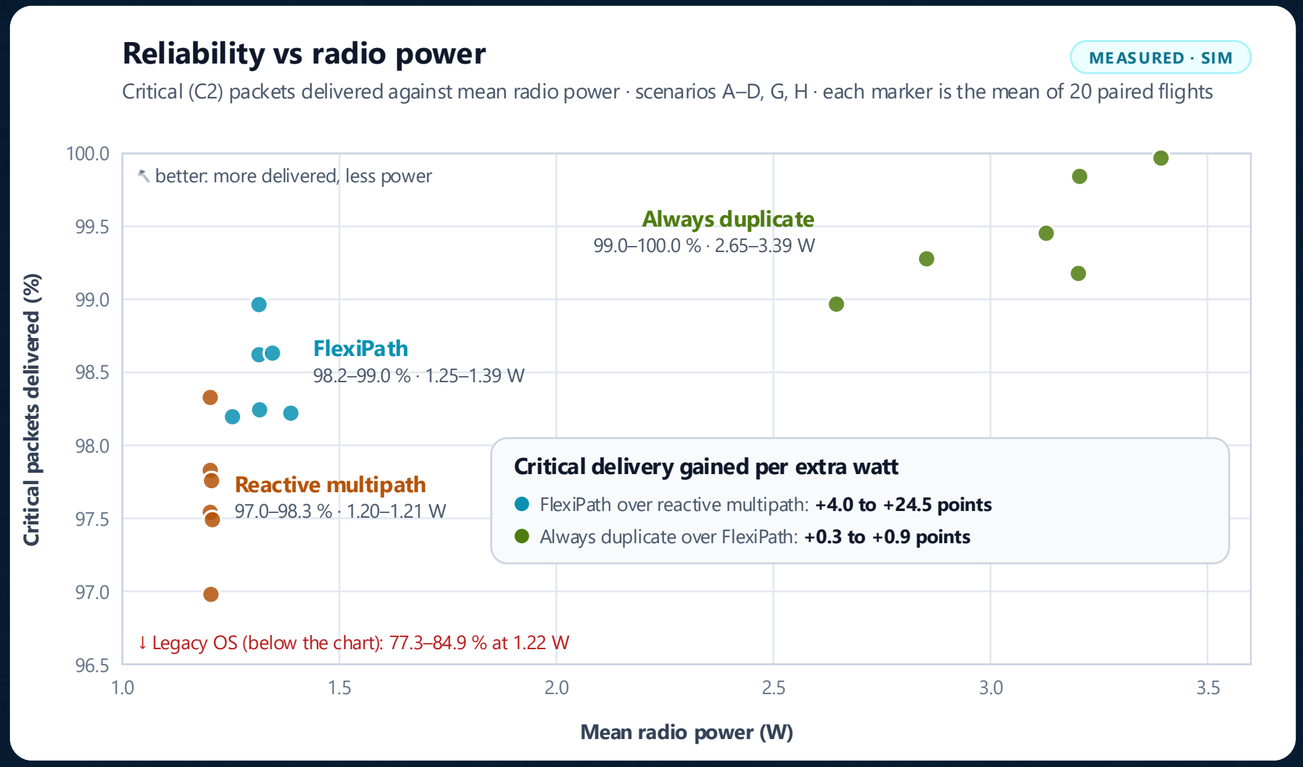
100% FlexiPath kept command and control in all scenarios with the piloted 300 ms rule

1.3–4.6 s FlexiPath acts before the path fails, versus 0.3–0.5 s after for reactive multipath

25–40% Fewer critical packets lost versus reactive multipath for about 0.1 W more

4.0–7.1 pp/W Reliability per extra watt versus reactive multipath **(24.5 in D)**

4.5–21× Less productive: always duplicate over FlexiPath delivers only **0.3–0.9 pp/W**



Reliability against radio power, scenarios A–D, G and H (paired means). FlexiPath sits between the two existing designs: 0.6–1.2 points more critical delivery than reactive multipath for 0.05–0.18 W more, and 2.1–2.6× less power than always duplicate.

Flights that kept C2 under the piloted 300 ms rule (Legacy · Reactive multipath · Always duplicate · FlexiPath): A, B, C 0 · 100 · 100 · 100 — D 0 · 0 · 100 · 100 — G, shared radio shadow 0 · 0 · 40 · 100 — H, map out of date 0 · 0 · 100 · 100.

Robustness: the ablations show trend prediction carries the gain (the burst adds no measurable reliability in-model); the results hold on held-out seeds; under the 1 s waypoint rule every multipath stack keeps C2, so the piloted-rule advantage needs a fast-resume spare modem.

Guidelines for QoS and QoE across heterogeneous networks (criterion 05)

01	Anchor the session, not the interface – a **fixed relay address** removes every reset.
02	Act on the trend, not on the failure – a **3 s signal trend** gives seconds of warning.
03	Prepare the next path just in time, and only then – warm-probe at **1.8 W**, do not keep radios on.
04	Duplicate only what needs it, only around the handover – a **400 ms burst** costs ≈ 1 % of protected bytes.
05	Deliver the **first copy**; never make real-time traffic wait in order.
06	Detect a dead path faster than the application can tolerate – within the **300 ms C2** budget.
07	**Hold still** unless moving clearly pays – a reversed trend releases the prediction at no cost.
08	Measure what **users feel**, on matched pairs – interruption, delivery, MOS, power, on identical seeds.
09	Use what you know about the route – a **coverage map** beats any reaction when two radios die together.



Plan for real-life use, limits, and declaration

Deployment path, proven limits, and our AI-use declaration

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DEPLOYMENT

A userspace agent on the vehicle's companion computer (Linux) controls modems via standard interfaces.
The anchor relay sits at the operator's ground station or cloud edge.
It runs under existing C2 and video applications; nothing changes.

LIMITS

All results are simulated.
The radio model is evaluated at ground level.
Piloted rule needs a spare modem that resumes in ~**100** ms (5G RRC_INACTIVE or connected-mode DRX).
At **260 ms**, an unexpected outage still forces a hover.

REAL-LIFE USE

Phase 1 (bench): laptop with Wi-Fi and a 5G dongle runs the included handover probe to measure one real network change. **Phase 2** (field): test drone with Wi-Fi, 5G and a LEO terminal on a surveyed corridor; first survey flights build the coverage map. **Phase 3** (pilot): one operator, one corridor, the paired-comparison method of this study applied to real flights.

AI-USE DECLARATION

AI tools (Anthropic Claude) assisted with code, analysis drafting, this deck and the video; its narration is an AI voice (Kokoro).
The design decisions, the experiments and their interpretation are the team's own, and every member can explain their part in their own words.